

Dr. Kevin Decker

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Computer Vision Engineer specializing in multimodal perception systems bridging research and production ML

EXPERIENCE

Staff AI/ML Researcher

Lockheed Martin | Littleton, CO | June 2025 – Present

- Served as AI/ML Team Lead for an \$11M program, driving architecture, technical direction, and delivery across a 7-engineer computer vision team while remaining hands-on in model and infrastructure development.
- Led development of next-generation SAR object detection systems using DETR variants, incorporating SAR-specific SSL approaches including SIMMIM and SAR-JEPA with Swin Transformer backbones.
- Architected a modality-agnostic ML platform enabling rapid experimentation, evaluation, and deployment across sensors, tasks, and model architectures, replacing a legacy research codebase.
- Architected and deployed a scalable object detection metrics framework replacing pycocotools entirely, enabling support for standard benchmarks and AFRL ATR/IC-specific metrics across multiple program.
- Developed scalable iSpy API tooling and data infrastructure for large-scale SAR data interrogation, curation, and analysis, enabling efficient management and exploitation of thousands of SAR image.

Senior AI/ML Researcher

Riverside Research | Fairborn, OH | June 2022 – June 2025

- Pioneered the development of neural network models leveraging 3D point-convolutions for spatial-temporal analysis of picosecond-duration events from single-photon counting camera data.
- Designed and implemented a 3D trajectory prediction algorithm using temporally directed graphs and Dijkstra's algorithm, achieving state-of-the-art performance.
- Fine-tuned Detection Transformer (DETR) for object detection on panchromatic imagery, employing transfer learning to significantly improve mAP and reduce false positives over the existing solution.
- Applied Meta's Segment Anything model for multi-object detection, image segmentation, and tracking in full-motion video.
- Contributed to the development of a neural network regression model for predicting the motion trajectory of objects from radar-based imagery. Used CycleGAN to transfer real-world data style to simulated data, increasing useable data for training.

AI/ML Researcher

Defense Engineering Corp. | Beavercreek, OH | January 2020 – June 2022

- Conducted research on multimodal fusion architectures, developing pixel and point convolution-based neural networks for semantic segmentation of lidar and hyperspectral imagery.
- Contributed to the development of a multimodal EO and lidar fusion network for semantic segmentation of the Dayton Annotated Laser Earth Scan (DALES) dataset; results were published in the 2021 IEEE AIPR Workshop.
- Created an advanced image processing software suite for lidar systems, featuring modular frameworks for calibration, statistical image processing, and AI-enhanced analysis pipelines.

AI/ML Engineer

InfoSciTex | Beavercreek OH | September 2018 – January 2020

- Developed CNN-based automatic target recognition models for lidar data and integrated them into simulation pipelines for automated dataset generation; results published in the Military Sensing Symposium.

Software Engineer

UES Inc | Beavercreek OH | May 2015 – September 2018

- Designed and deployed the Autonomous Research System (ARES) for autonomous experimentation in carbon nanotube synthesis; published in *npj Computational Materials*.

EDUCATION

Ph.D. in Computer Science

Air Force Institute of Technology | WPAFB, OH | January 2017 – December 2024

- Dissertation: "Hyperspectral and Lidar Fusion and the Evaluation of Sampling Methodologies in Remote Sensing"

MSc. in Computer Science

BA in Computer Science

University of Cincinnati | Cincinnati, OH | May 2009 – May 2015

PUBLICATIONS

Earlier works on request, or LinkedIn profile

- K. T. Decker and B. J. Borghetti. "Composite Style Pixel and Point Convolution-Based Deep Fusion Neural Network Architecture for the Semantic Segmentation of Hyperspectral and Lidar Data". Remote Sensing 2022, Vol. 14, Page 2113, 14(9):2113, apr 2022.
- K. T. Decker and B. J. Borghetti. "Hyperspectral Point Cloud Projection for the Semantic Segmentation of Multimodal Hyperspectral and Lidar Data with Point Convolution-Based Deep Fusion Neural Networks". Applied Sciences 2023, Vol. 13, Page 8210, 13(14):8210, jul 2023.
- K. T. Decker and B. J. Borghetti. "A survey of sampling methods for hyperspectral remote sensing: Addressing bias induced by random sampling". Remote Sensing, 2025.

Technologies: Python, PyTorch, OpenMMLab, Plotly, Docker/Podman, Linux, Git, LaTeX, AWS (EC2, S3, EFS, FSx)